Name: Dulce Renteria

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Instructor: Samuel Chukwuemeka.

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Objectives:

1. Verify the check digit for ISBN-10
2. Verify the check digit for ISBN-13
3. Convert ISBN-10 to ISBN-13
4. Convert ISBN-13 to ISBN-10

Calculations:

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ISBN-10: 0134954696 (Modulo 11)

Objective #1:

Given: 0134954696

Approach: Multiplication-Modulo Approach

$$0 \cdot 1 = 1$$

$$1 \cdot 2 = 2$$

$$3 \cdot 3 = 9$$

$$4 \cdot 4 = 16$$

$$9 \cdot 5 = 45$$

$$5 \cdot 6 = 30$$

$$4 \cdot 7 = 28$$

$$6 \cdot 8 = 48$$

$$9 \cdot 9 = 81$$

Total of the sum

$$0 + 2 + 9 + 16 + 45 + 30 + 28 + 48 + 81 = 259$$

Divide the product by modulo 11

Reminder =259 modulo 11=6

The check digit is **6**

Objective #2

Given: 978-0134954 691

Approach: Multiplication-Modulo-Subtraction Approach

$$9 \cdot 1 = 9$$

$$7 \cdot 3 = 21$$

$$8 \cdot 1 = 8$$

$$0 \cdot 3 = 0$$

$$1 \cdot 1 = 1$$

$$3 \cdot 3 = 9$$

$$4 \cdot 1 = 4$$

$$9 \cdot 3 = 27$$

$$5 \cdot 1 = 5$$

$$4 \cdot 3 = 12$$

$$6 \cdot 1 = 6$$

$$9 \cdot 3 = 27$$

Sum of the products

$$9 + 21 + 8 + 0 + 1 + 9 + 4 + 27 + 5 + 12 + 6 + 27 = 129$$

Sum divided by modulo

$$129 \div 10 = 12$$

remainder of 9

Modulo minus remainder

$$10 - 9 = 1$$

The check digit is 1

Objective 3: Convert ISBN-10 to ISBN-13

Given number: 0134954696

Step 1: Remove the check digit: 013495469

Step 2: Append 978: 9780134954689

Step 3: Calculate the check digit in the second objective, which is 1

ISBN-13 is 9781349546891

Objective 4: Convert ISBN-13 to ISBN-10

Given number: 978-0134954691

Step 1: Remove the check digit: 978013495469

Step 2: Remove the number 978: 013495469

Step 3: Calculate the check digit in the first objective, which is 6

ISBN-10 is 0134954696

References (APA7):

Charles, R. I., Berry, R. Q., Champagne, Z., Schielack, J. F., Wray, J. A., & Fennell, F. (2020). *Envision mathematics. common core*. Pearson.

Check digits for International Standard Book Number (ISBN). Samuel Dominic Chukwuemeka. (n.d.). https://mathematicseducation.appspot.com/NumberTheoryProjects/Projects-ISBN.html